GAJENDRA MANDAL

gajendramandal2003@gmail.com ◇ +91-6264769759 ◇ linkedin ◇ Github

SUMMARY

B.Tech (Hons.) Data Science graduate from CSVTU, Bhilai, with all academic requirements completed and degree expected in July 2026. My work spans machine learning, computer vision, thermal imaging, and IoT systems, with hands-on experience turning research into working, end-to-end solutions. At IIT Gandhinagar, I built thermal imaging–based respiration monitoring pipelines and applied vision-language models to thermal data, sharpening my skills in model development and applied research. I'm looking to bring this experience to real-world data science and AI challenges in a team that values building things that actually work.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech(H) CSE(DS)	Chhattisgarh Swami Vivekanand Technical University	8/10(CPI)	2022-2026
Senior Secondary (12th Board)	Kendriya Vidyalaya, Ambikapur	83%	2022
Secondary (10th Board)	Kendriya Vidyalaya, Ambikapur	65%	2020

WORK EXPERIENCE

AI/ML Intern **March 2026 – Present**
Stembotix, Gandhinagar.

- Working on IoT and AI/ML projects, including sensor-integrated dashboards and real-time data pipelines for embedded systems.
- Developing and deploying full-stack IoT solutions using ESP32/Arduino hardware, Node.js, PostgreSQL, and multi-protocol communication (MQTT, WebSocket, HTTP).
- Contributing to system design, firmware debugging, and cloud deployment workflows for scalable IoT infrastructure.

Research Intern – Computer Vision **Feb 2025 – July 2025**
Indian Institute of Technology Gandhinagar.

- Developing thermal imaging–based respiration and sleep-apnea detection pipelines, including YOLO-based nostril detection and vision–language models (VLMs) for thermal data interpretation.
- Utilizing Python, PyTorch, OpenCV, and Docker for image processing, model training, and deployment within research workflows.
- Contributed to improved respiration signal extraction accuracy and dataset annotation automation, enabling scalable analysis of large thermal datasets for healthcare AI research.

PROJECTS

Screen Apnea – Thermal-Based Respiration Monitoring System

- Developed a non-contact respiration monitoring system using thermal imaging to detect shallow breathing and apnea-related anomalies without wearable sensors.
- Trained and optimized **YOLOv8 models** for real-time nostril detection, achieving mAP@0.5 of 0.978 and enabling edge-device compatible inference.
- Implemented a signal processing pipeline (mean pixel extraction, downsampling, bandpass filtering) and validated respiration frequency against chest belt ground truth (MAE: 0.31 Hz).

- Leveraged deep learning frameworks to experiment with multiple object detection models and applied various signal filtering techniques for noise reduction and smoothing.

Real-Time Threat Detection System

- Built a real-time surveillance framework using YOLOv8 to detect guns, knives, and blunt objects in public-safety scenarios.
- Curated a custom 7,959-image dataset (336 self-collected + 7,623 public) across three threat classes for model training.
- Achieved improved recall and precision for the blunt-object class through extended training, with no signs of overfitting.
- Designed the system for real-world deployment in campuses, public spaces, and transportation areas.

SKILLS

- **Programming Languages:** Proficient in C, C++, Python, R, SQL.
- **Data Science & Machine Learning:** DSA, Pandas, NumPy, Matplotlib, Scikit-learn, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, YOLOv8, PyTorch, OpenCV.
- **Web Development:** HTML, CSS, JavaScript, Node.js, Express.js, MongoDB.
- **IoT & Embedded Systems:** ESP32, ESP-01, Arduino Nano/Uno, Sensor Integration, MQTT, WebSocket, Blynk, ThingSpeak.
- **Tools & Platforms:** Git/GitHub, WordPress, VS Code, MATLAB, Google Colab, Jupyter Notebook, MS Office Suite (Excel, PowerPoint, Word).
- **Creative Skills:** DaVinci Resolve (Video Editing), Canva (Graphic Designing).
- **Soft Skills:** Resource and time Management, problem-solving, teamwork, communication and leadership.

PUBLICATIONS

- **Real-Time Threat Detection from Surveillance Cameras Using Machine Learning**
  First Author, Accepted at the 2026 IEEE International Conference on Recent Advances in Electronics, Communication, Computing, Automation, and Power (RECCAP 2026).
  Preprint: http://arxiv.org/abs/2606.05708

ACHIEVEMENTS

- **Hackathons:** Participated in the National-Level Data Analytics Hackathon organized by IIM Nagpur, Smart India Hackathon (SIH) 2025, and multiple hackathons organized by the Coding Club.
- **IDC-2024:** Selected to represent NSS and our university at the Independence Day Ceremony (IDC) 2024 in New Delhi.
- **Rajya Puraskar:** Awarded for outstanding performance and dedication in Scouts and Guides.

VOLUNTEERING

National Service Scheme

- ACM Summer School–IIT Gandhinagar (2025): Assisted in coordination of technical sessions and event logistics.
- Served as Head Boy of National Service Scheme (NSS), demonstrating leadership and organizational skills.
- Achieved B-Certificate in the session 2023-24 reflecting commitment and dedication to NSS activities.
- Organized and coordinated multiple student branch events while actively participating in national-level camps, contributing to leadership development and community engagement.